



Professional Virtual Lab Environment

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Table Of Contents:

- Introduction
- Objective
- Theory Topics
 - Introduction to MSCE
 - Role of a System Administrator
 - Cloud Computing Fundamentals
 - Virtualization Technologies
 - Hypervisors (Type-1 vs Type-2)
 - VMware Workstation Overview
 - Windows 10 Architecture
 - Windows Installation Process
 - Introduction to Windows Disk Management
 - Basic Disk Part CLI Commands
- Software Requirements
- Implementation
 - VMware Installation
 - Virtual Machine Creation
 - Windows 10 Installation
 - Disk Management using Command Line
 - Complete Windows Installation
- Challenges faced
- Learning outcomes
- Conclusion
- References

Introduction

A reliable and well-structured virtual lab environment is an essential tool for every professional System Administrator. It provides a safe and controlled environment for learning, testing, configuring, and troubleshooting operating systems and enterprise infrastructure without affecting physical machines. Through virtualization technologies, multiple operating systems can be deployed and managed efficiently on a single computer, making it an indispensable part of modern IT and system administration.

Objective

This document provides a step-by-step guide for creating a professional Windows 10 virtual lab environment using VMware Workstation. The objective is to demonstrate the process of installing and configuring a virtual machine, performing disk management using DiskPart, and documenting the complete deployment.

Theory Topics

Introduction to MSCE

Microsoft Certified Solutions Expert (MCSE) was a certification that validated expertise in designing, implementing, and managing Microsoft technology solutions. While Microsoft no longer offers the MCSE certification, it represented a high level of skill and knowledge across various IT domains such as server infrastructure, cloud platforms, and

data management. It was commonly used to demonstrate proficiency and advance careers in IT roles requiring in-depth knowledge of Microsoft products.

Microsoft retired the MCSE certification and replaced it with role-based certifications that better reflect modern IT job roles. Unlike MCSE, which focused on Microsoft products, the new certifications emphasize practical skills in areas such as cloud computing, cybersecurity, DevOps, data management, and system administration.

Microsoft now offers certifications for various roles and experience levels, including expert certifications such as:

- Microsoft 365 Certified: Administrator Expert
- Microsoft Certified: Azure Solutions Architect Expert
- Microsoft Certified: Cybersecurity Architect Expert
- Microsoft Certified: DevOps Engineer Expert

Each certification follows a specific learning path with its own prerequisites.

Role of a System Administrator

System administrators are normally tasked with the installation, maintenance, configuration and repair for servers, networks and other computer systems within an organization. That means you're directly supports network operations, sometimes working directly inside the NOC (the network operations center).

Working tirelessly, often outside of regular working hours, SysAdmins ensure that the systems within an organization are up and available with minimal downtime, both for users and the systems themselves. Unless something goes wrong, the work of a Systems Administrators typically goes unnoticed by those around them.

System administrator roles typically include:

1. Server/Web administrator: Responsible for the installation, configuration, and maintenance of servers within an organization.
2. Network administrator: Responsible for overseeing the organization's network system, identifying potential bottlenecks, implementing robust network security measures to protect against cyber threats, and troubleshooting any network-related issues that may arise.
3. Database administrators: Responsible for designing and configuring databases and data management software to suit the specific needs and requirements of your organization's data. They ensure data is stored, accessed, and processed in a secure and efficient manner.

Cloud Computing Fundamentals

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like Amazon Web Services (AWS).

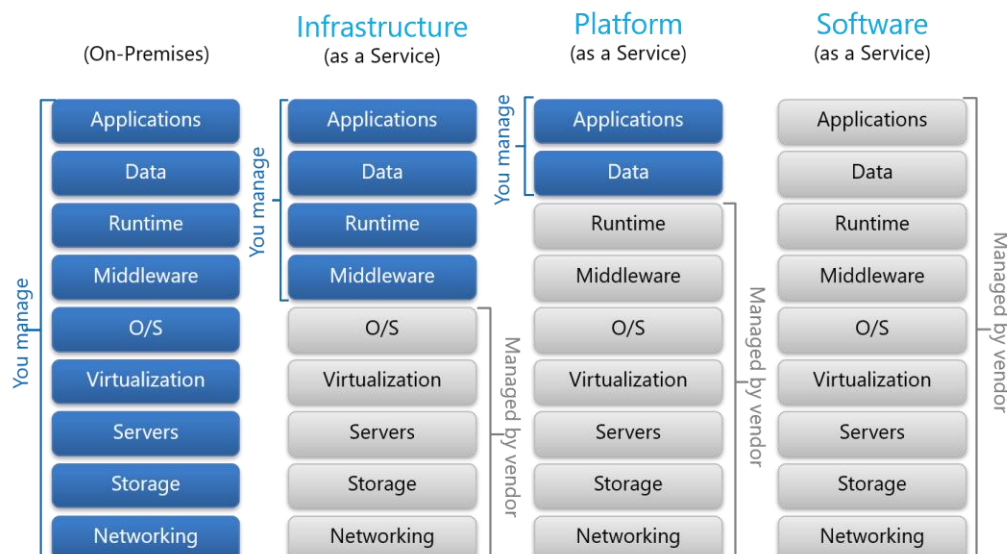
Some benefits of cloud Computing:

- Agility: You can deploy technology services in a matter of minutes, and get from idea to implementation several orders of magnitude faster than before.
- Elasticity: you don't have to over-provision resources up front to handle peak levels of business activity in the future. Instead, you provision the amount of resources that you actually need.

- **Cost savings:** The cloud allows you to trade fixed expenses (such as data centers and physical servers) for variable expenses, and only pay for IT as you consume it.

Cloud Computing Models:

- **SaaS (software-as-a-service):** Provides a full application stack as a service that customers can access and use. SaaS solutions often come as ready-to-use applications, which are managed and maintained by the cloud service provider.
- **IaaS (Infrastructure as a service):** Delivers on-demand infrastructure resources, such as compute, storage, networking, and virtualization. With IaaS, the service provider owns and operates the infrastructure, but customers will need to purchase and manage software, such as operating systems, middleware, data, and applications.
- **PaaS (Platform as a service):** Delivers and manages hardware and software resources for developing, testing, delivering, and managing cloud applications. Providers typically offer middleware, development tools, and cloud databases within their PaaS offerings.



Virtualization Technologies

Server virtualization is the process of dividing a physical server into multiple unique and isolated virtual servers by means of a software application. Each virtual server can run its own operating system independently. This technology masks server resources from server users, including the number and identity of operating systems, processors, and individual physical servers.

Server virtualization abstracts network, storage, and compute resources so applications, services, and functions become less dependent on physical hardware. IT administrators can provide applications and services with their own environment, including an operating system, support software, network, and storage resources, that is less prone to problems caused by other workloads running on the same underlying resources.

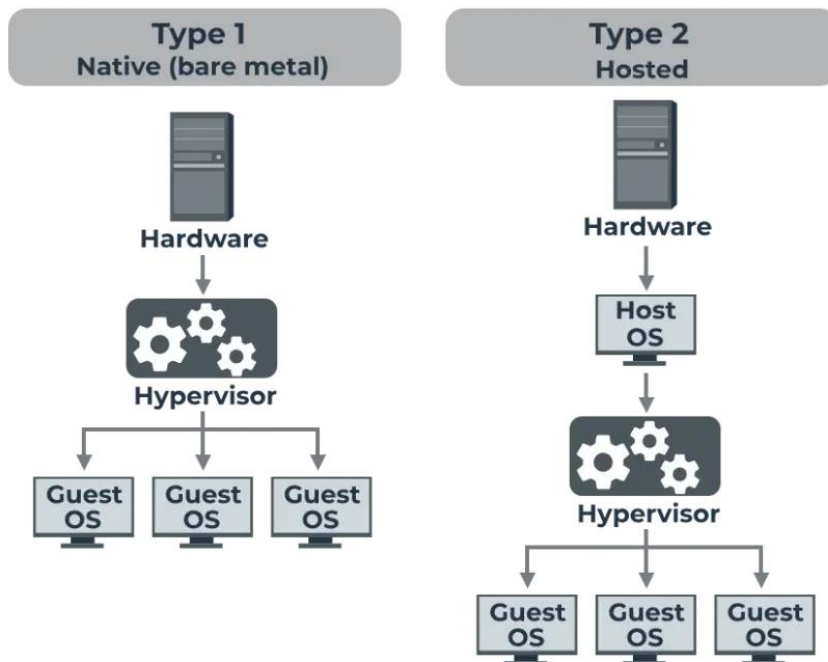
Hypervisors (Type-1 vs Type-2)

At the heart of most virtualization implementations is the hypervisor (also called a Virtual Machine Monitor, or VMM), which serves as the key component that creates and manages multiple virtual machines on a single physical host. The hypervisor forms a layer between the hardware and the operating systems, performing critical functions.

There are two main hypervisor types:

1. Type 1, or “bare metal” : runs straightforwardly on the physical hardware of the host machine.
2. Type 2, or “hosted” : usually installed on top of an accessible operating system.

Hypervisor or Virtual Machine Monitor (VMM)



VMware Workstation Overview

VMware Workstation is a Type-2 hypervisor that enables multiple isolated virtual machines to run on a single physical computer. As a conventional software application, it also generates standard system artifacts during installation and use, including DLL files, Prefetch entries, and temporary files, making its presence readily identifiable.

Windows 10 Architecture

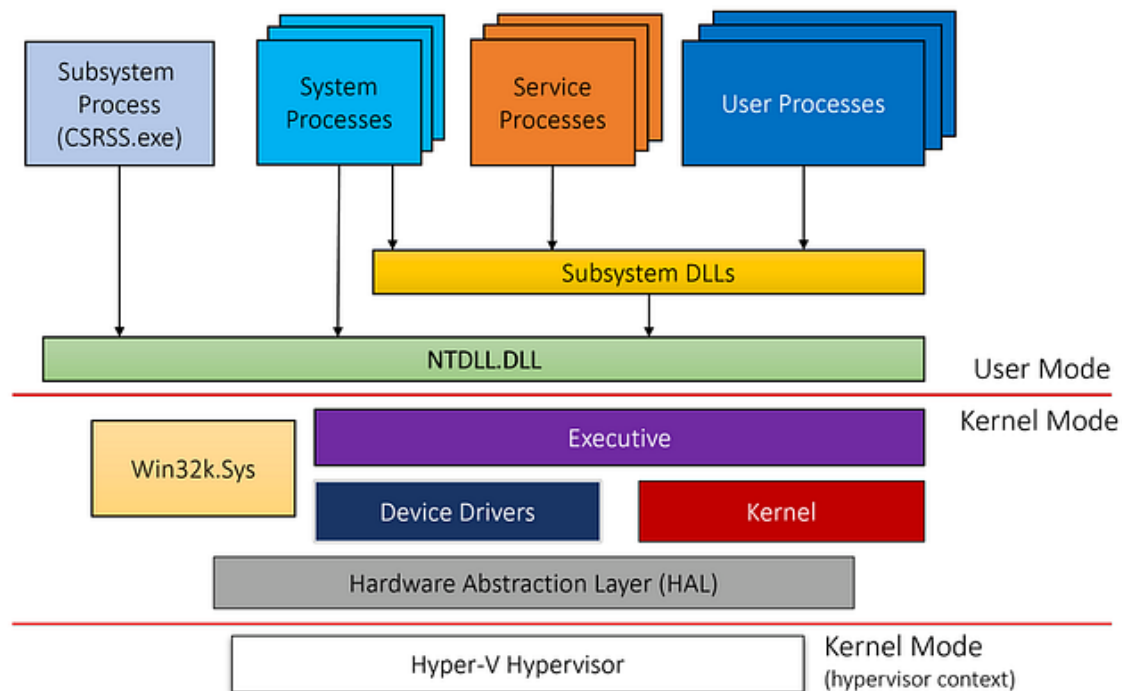
The Windows architecture can be broadly divided into two main modes: User Mode and Kernel Mode. This separation is crucial for system stability and security.

The reason Windows uses only two levels is that some hardware architectures, such as ARM today and MIPS/Alpha in the past, implemented only two privilege levels. Settling on the lowest minimum bar allowed for a more efficient and portable architecture, especially as

the other x86/x64 ring levels do not provide the same guarantees as the ring 0/ring 3 divide.

User mode is a restricted environment where most applications execute. It provides a safe and isolated space for programs to run, preventing them from directly accessing critical system resources or interfering with other applications.

Kernel mode is the privileged execution mode where the core operating system components reside. It has direct access to the system's hardware and memory. This mode is responsible for managing system resources, enforcing security policies, and providing services to user-mode applications.



Windows Installation Process

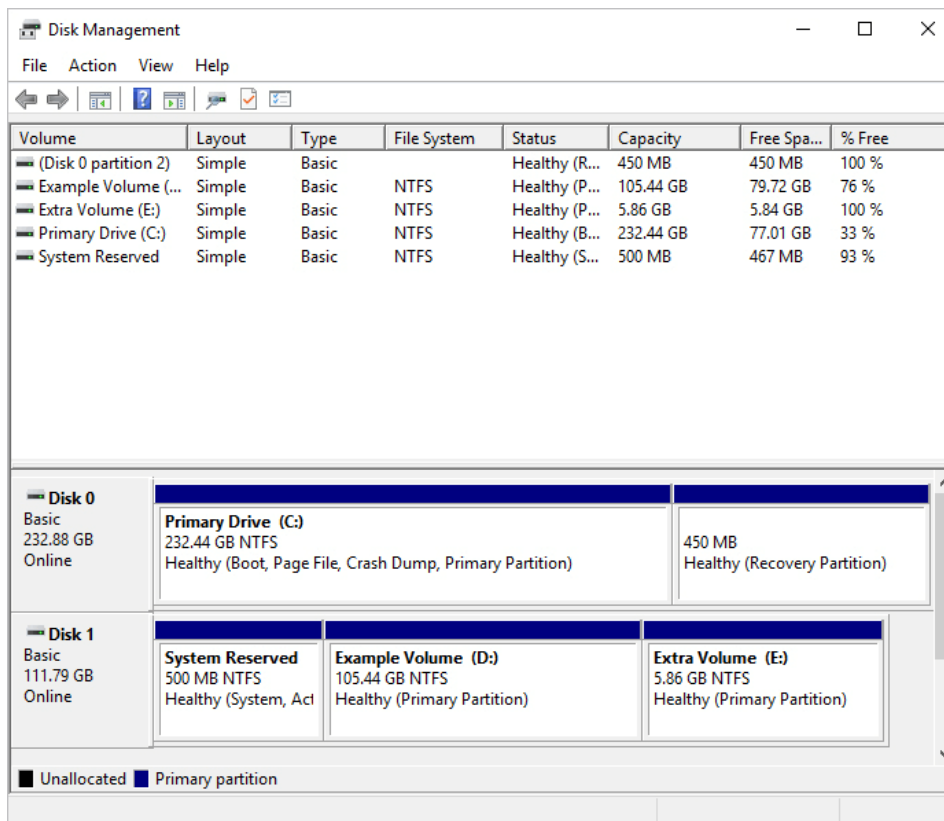
The Windows Installation Process is the procedure of deploying the Windows operating system onto a computer or virtual machine. It

prepares the system by copying the necessary operating system files, configuring hardware settings, and performing the initial system setup.

The installation process typically consists of the following stages: Boot from the USB Drive, Begin the Installation, Choose Your Edition, Agree to the License Terms, Select the Drive, Wait for the Installation, Complete the Setup, and Activate.

Introduction to Windows Disk Management

Disk Management is a system utility in Windows for advanced storage operations. You can use the utility to see information about each drive on your computer and all partitions in each drive. The utility shows partition details, including statistics and the amount of space allocated or used.



Basic Disk Part CLI Commands

The diskpart command interpreter helps you manage your computer's drives (disks, partitions, volumes, or virtual hard disks), it looks similar to the Disk Management tool, but the only difference is that diskpart is a command prompt utility which makes it more powerful and provides more functionality than the Disk Management tool.

Syntax:

diskpart <parameter>, command examples you can run include:

ACTIVE	- Mark the selected partition as active.
ADD	- Add a mirror to a simple volume.
ASSIGN	- Assign a drive letter or mount point to the selected volume.
ATTRIBUTES	- Manipulate volume or disk attributes.
ATTACH	- Attaches a virtual disk file.
AUTOMOUNT	- Enable and disable automatic mounting of basic volumes.
BREAK	- Break a mirror set.
CLEAN	- Clear the configuration information, or all information, off the disk.
COMPACT	- Attempts to reduce the physical size of the file.
CONVERT	- Convert between different disk formats.
CREATE	- Create a volume, partition or virtual disk.
DELETE	- Delete an object.
DETAIL	- Provide details about an object.
DETACH	- Detaches a virtual disk file.
EXIT	- Exit DiskPart.
EXTEND	- Extend a volume.
EXPAND	- Expands the maximum size available on a virtual disk.
FILESYSTEMS	- Display current and supported file systems on the volume.
FORMAT	- Format the volume or partition.
GPT	- Assign attributes to the selected GPT partition.
HELP	- Display a list of commands.
IMPORT	- Import a disk group.
INACTIVE	- Mark the selected partition as inactive.
LIST	- Display a list of objects.
MERGE	- Merges a child disk with its parents.
ONLINE	- Online an object that is currently marked as offline.
OFFLINE	- Offline an object that is currently marked as online.
RECOVER	- Refreshes the state of all disks in the selected pack. Attempts recovery on disks in the invalid pack, and resynchronizes mirrored volumes and RAID5 volumes that have stale plex or parity data.

Software Requirements

Hypervisor: VMware Workstation Pro / VMware Workstation Player.

OS Media: Windows 10 ISO (64-bit).

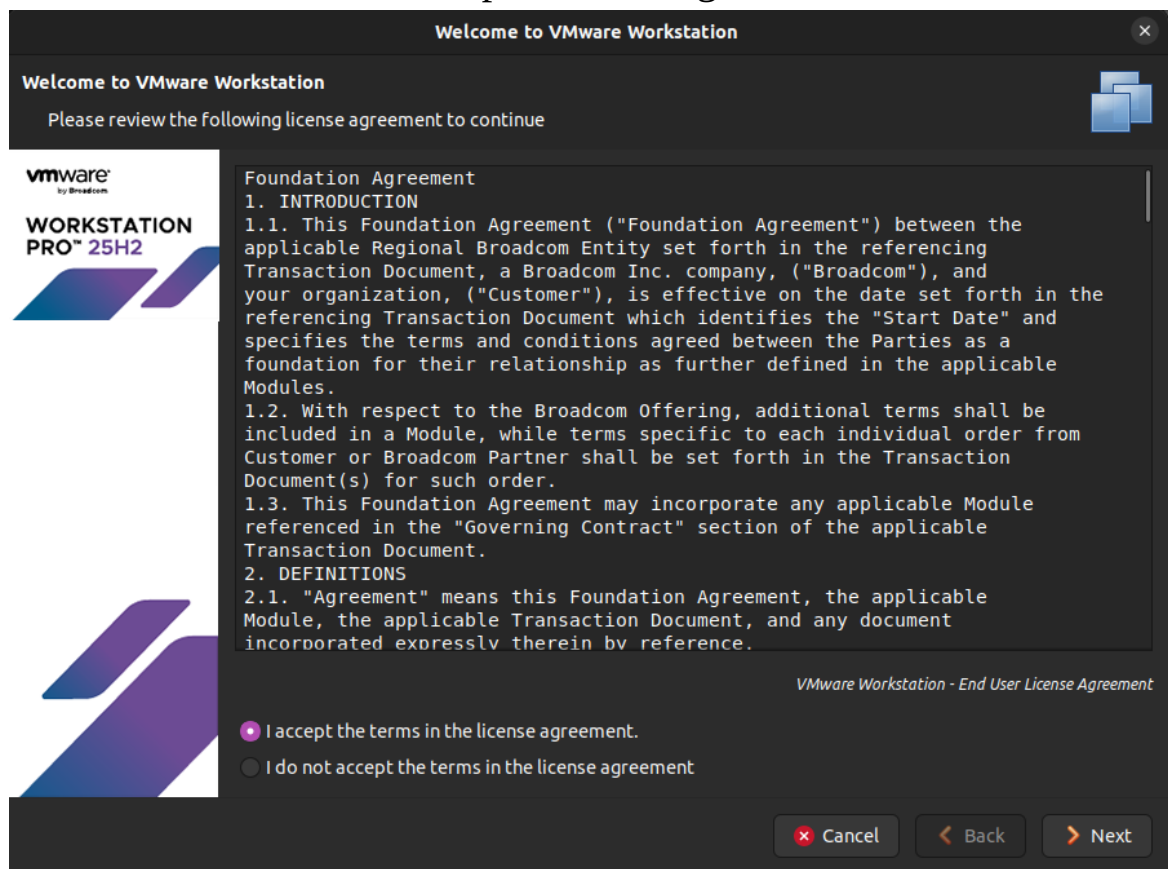
Hardware: Minimum 8 GB RAM (16 GB Recommended), Minimum 50 GB Free Disk Space.

Tools: Git (Optional but Recommended), Microsoft Word or PDF Editor for Documentation

Implementation

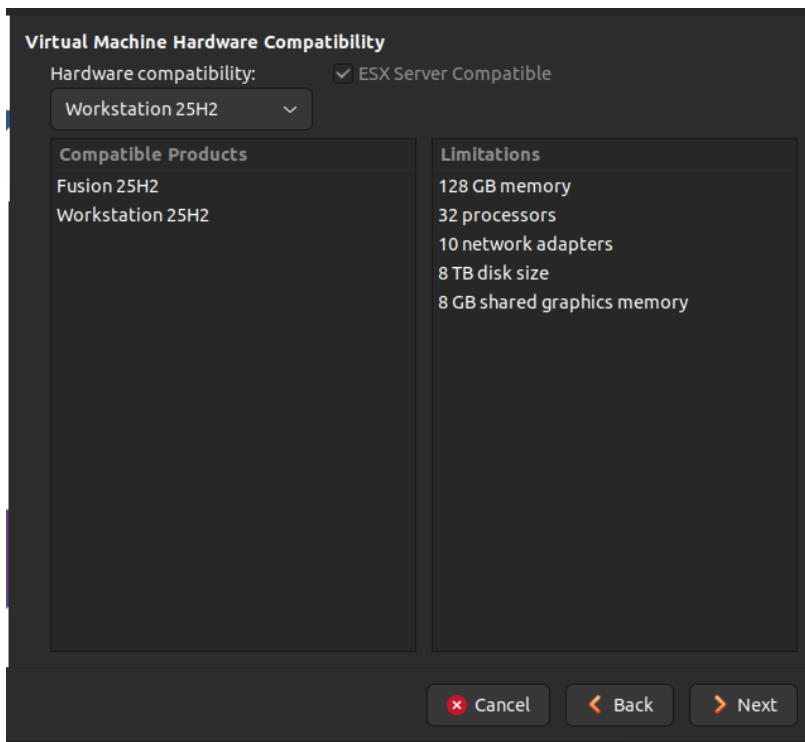
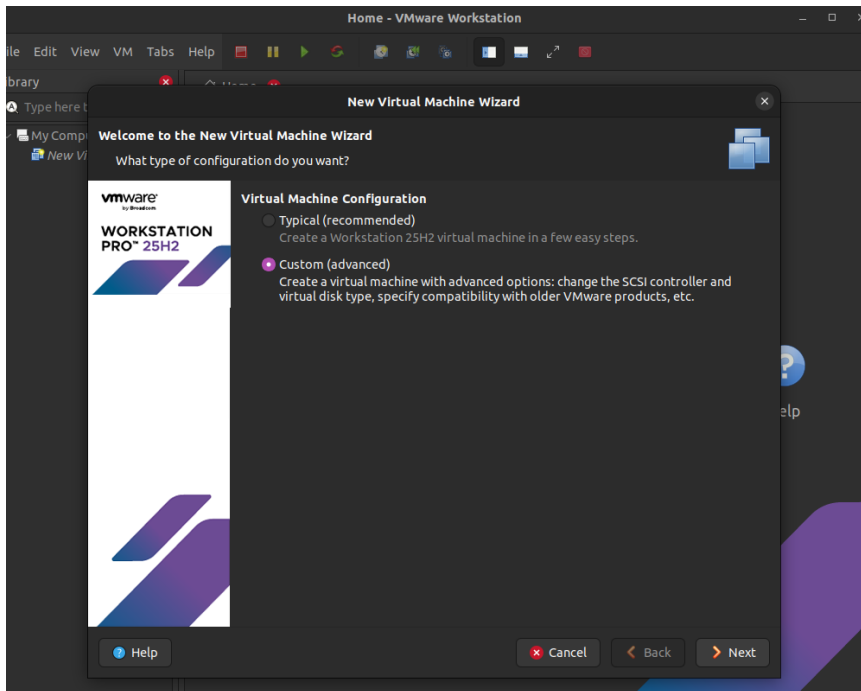
A. VMware Installation

Download, install, and accept terms of agreement

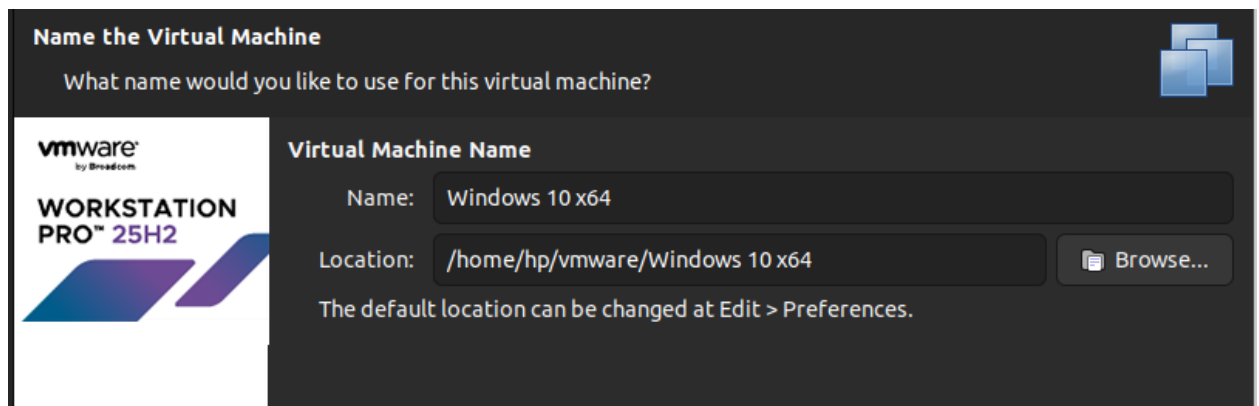
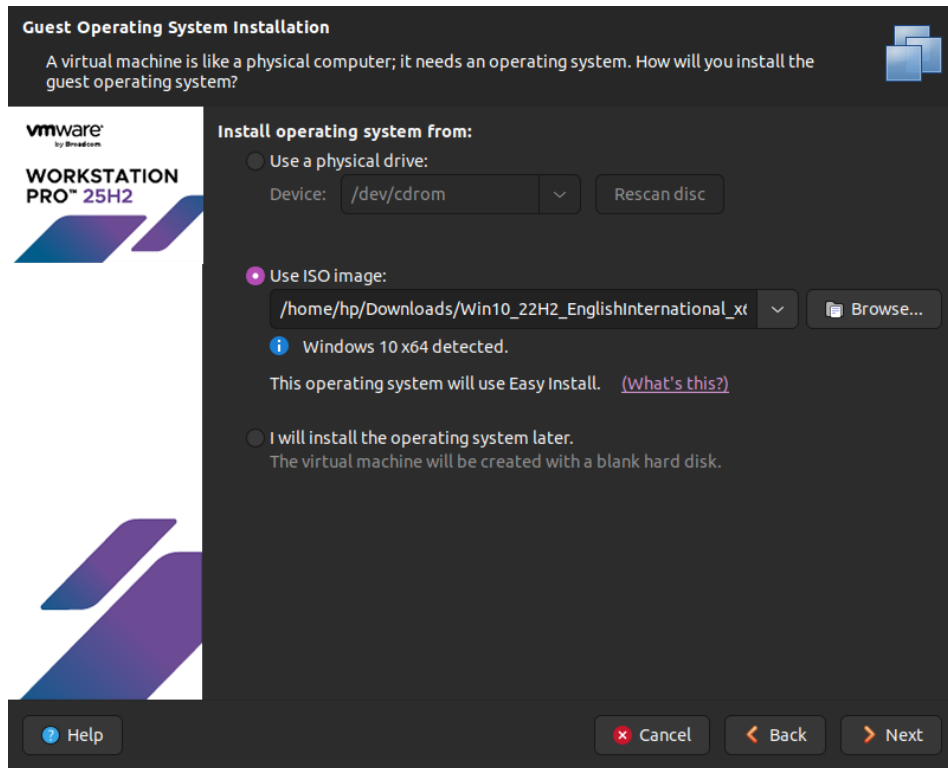


B. Virtual Machine Creation

Create a new VM, choose custom (advanced) -> default (Workstation 25H2)



-> select the windows10 ISO image



-> select UEFI for Modern Infrastructure Mirroring (BIOS simulating a legacy enterprise environment), and continue other setup (2 Cores, 4 GB RAM, NAT)

Firmware Type

What kind of boot device should this virtual machine have?

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WORKSTATION
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Firmware Type

- ☐ BIOS
- ☒ UEFI
- ☒ Secure Boot

Processor Configuration

Specify the number of processors for this virtual machine.

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PRO™ 25H2

Processors

Number of processors: 1

Number of cores per processor: 2

Total processor cores: 2

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Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 4096 MB

128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB

Maximum recommended memory
(Memory swapping may occur beyond this size)
13412 MB

Recommended memory
2048 MB

Guest OS recommended minimum
2048 MB



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Network Connection

- ☐ Use bridged networking
Give the guest operating system direct access to an external Ethernet network. The guest must have its own IP address on the external network.
- ☒ Use network address translation (NAT)
Give the guest operating system access to the host computer's Wi-Fi or Ethernet network connection using the host's IP address.
- ☐ Use host-only networking
Connect the guest operating system to a private virtual network on the host computer.
- ☐ Do not use a network connection



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Select I/O Controller Types

Which SCSI controller type would you like to use for SCSI virtual disks?

I/O Controller Types

SCSI controller: ☐ BusLogic ☐ LSI Logic ☒ LSI Logic SAS (Recommended) ☐ Paravirtualized SCSI

-  BusLogic SCSI controllers are not supported on 64-bit virtual machines.
-  LSI Logic SCSI controllers are not supported on Windows 10 x64 virtual machines.
-  Paravirtualized SCSI controllers are not supported on Windows 10 x64 virtual machines by Easy Install.



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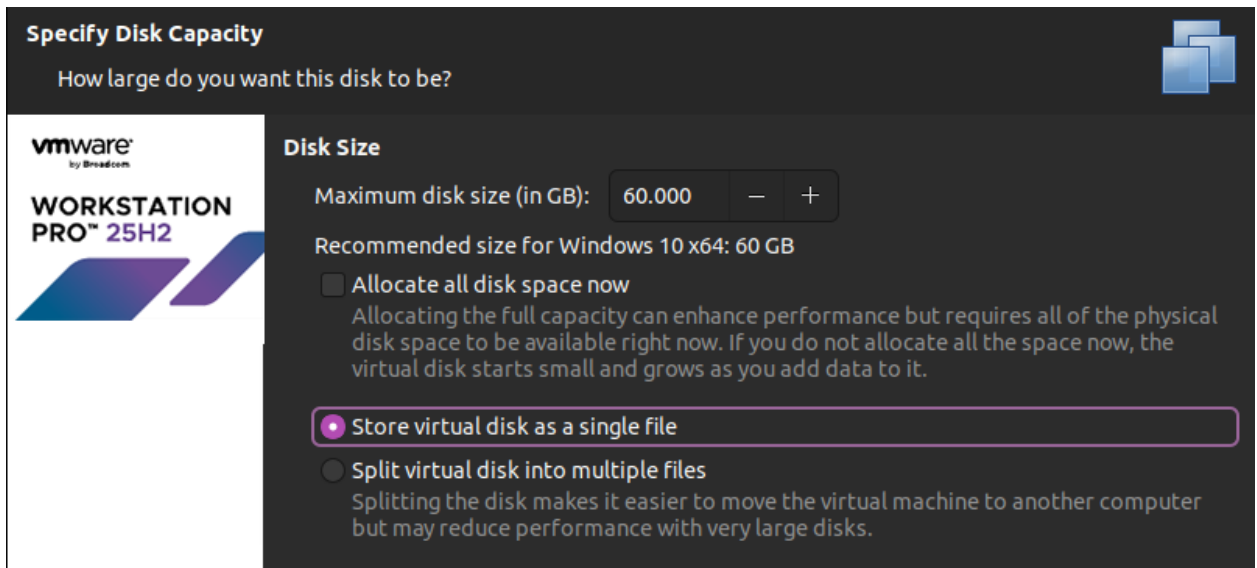
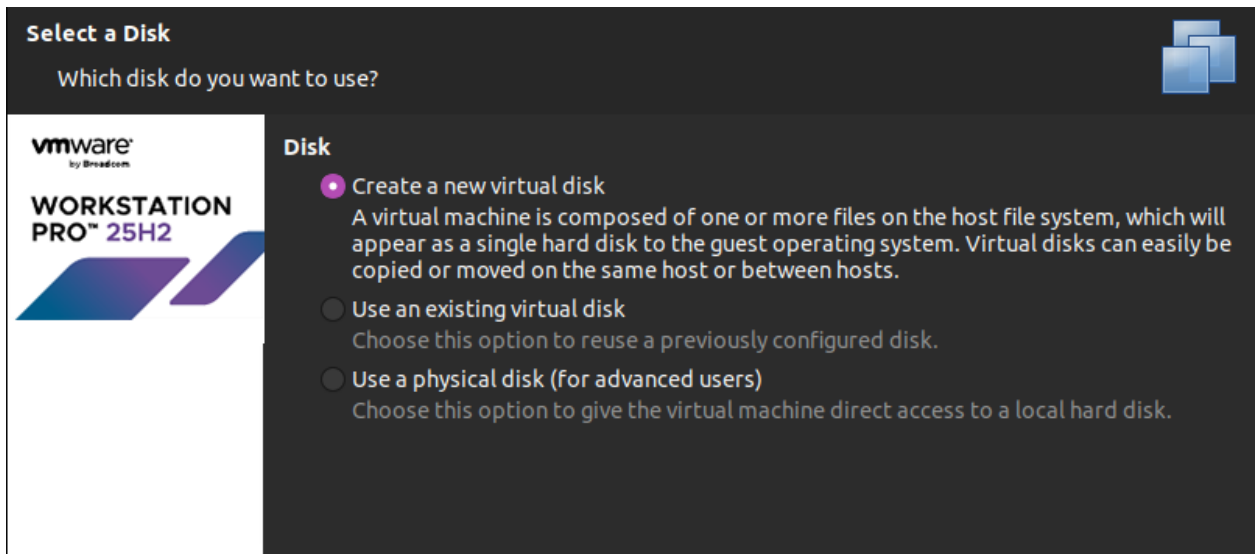
Select a Disk Type

What kind of disk do you want to create?

Virtual Disk Type

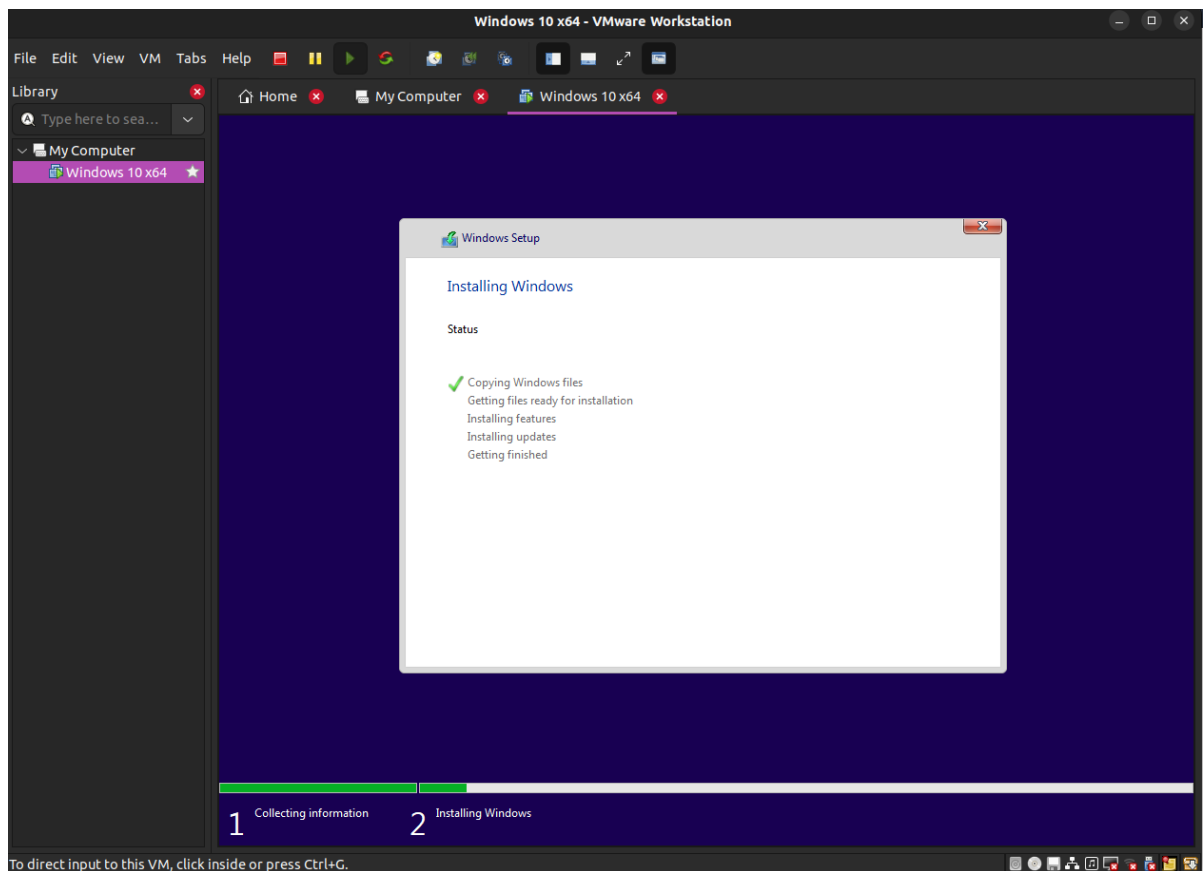
- ☐ IDE
- ☐ SCSI
- ☐ SATA
- ☒ NVMe (Recommended)

-> Create a new virtual disk keeps the environment safely isolated within a single virtual file

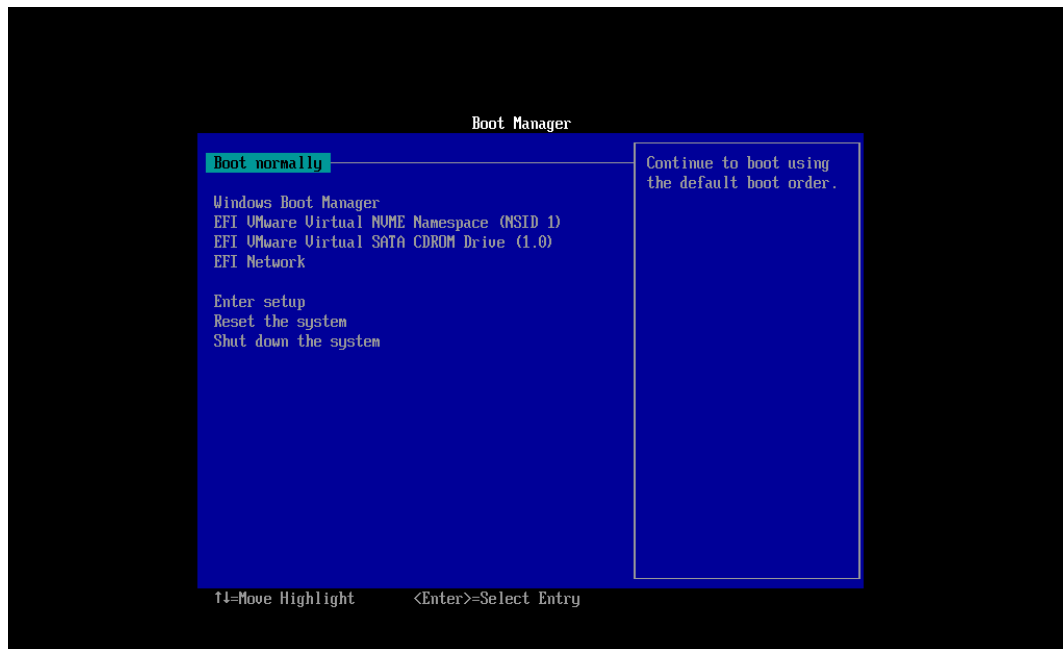




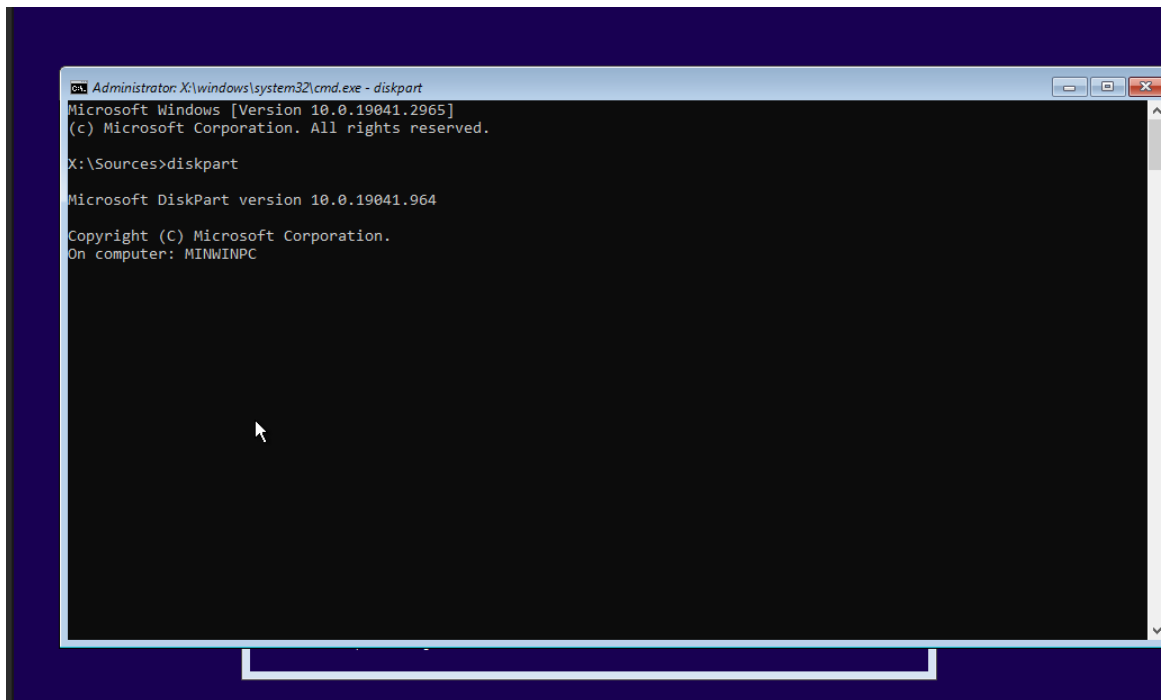
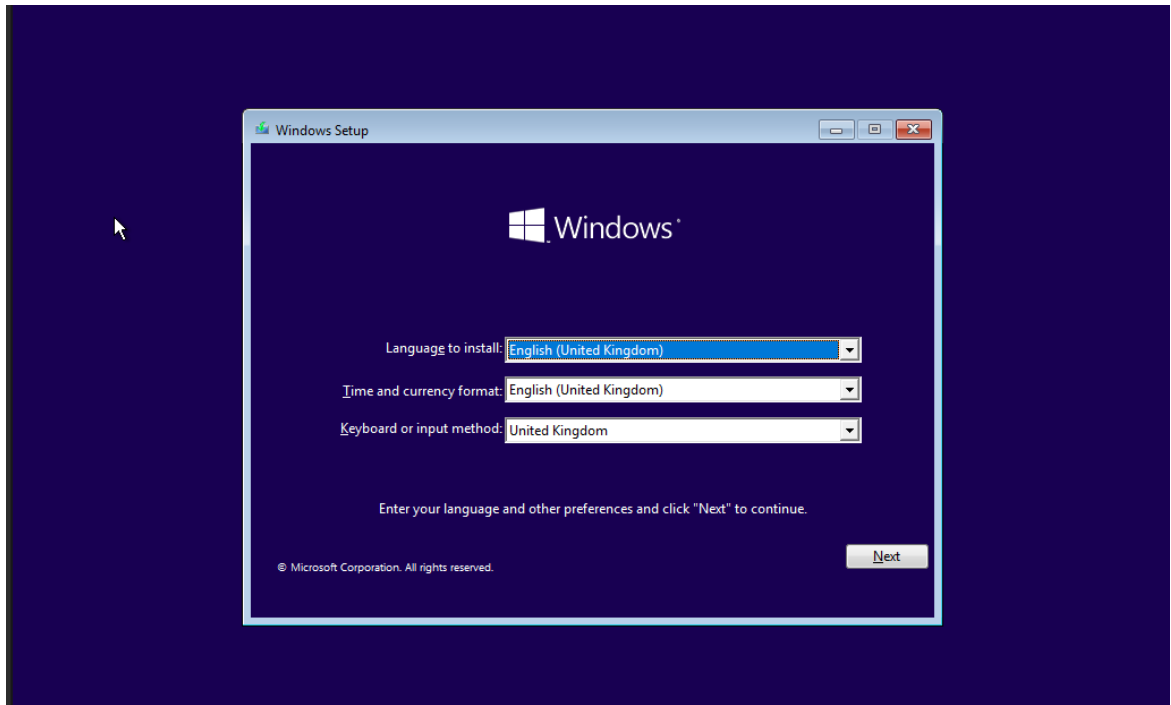
C. Windows 10 Installation



D. Disk Management using Command Line



- > select EFI VMware Virtual SATA CDROM Drive (1.0) and press Enter : this tells the VM to boot from your Windows ISO file
- > immediately after press spacebar -> then press shift + f10



Command execution:

- list disk : Displays all physical/virtual drives attached to the machine.

```
DISKPART> list disk
```

Disk ###	Status	Size	Free	Dyn	Gpt
Disk 0	Online	60 GB	0 B		*

- select disk 0: Shifts the command focus entirely onto your 60 GB virtual drive so the next partition rules apply specifically to it.

```
DISKPART> select disk 0
```

Disk 0 is now the selected disk.

- create partition primary size=21000 : carves out a Primary Partition of roughly 21 GB. This is the partition where your main Windows operating system files (C: drive) will eventually be installed.

```
DISKPART> create partition primary size=21000
```

DiskPart succeeded in creating the specified partition.

- create partition extended : Takes the remaining unallocated space left on the 60 GB drive and designates it as an Extended Partition wrapper. Extended partitions act as a container, allowing you to bypass old drive constraints and divide the leftover space into multiple sub-drives (logical drives).

Note: on GPT disk every partition is primary. On MBR there's distinction between primary and extended partitions, but MBR only works on legacy BIOS systems and not modern EFI systems

```
DISKPART> create partition extended  
DiskPart succeeded in creating the specified partition.
```

Figure 1: MBR system with extended partitions

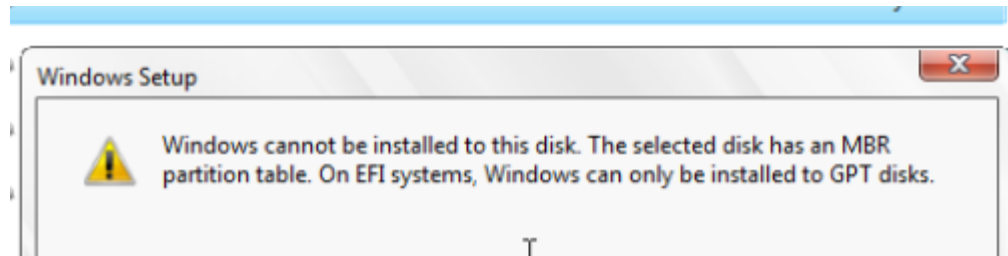


Figure 2: MBR

```
DISKPART> create partition extended  
Logical and extended partitions cannot be created on a GPT disk.  
Instead create primary partitions on a GPT disk.
```

Figure 3: using GPT

- create partition logical size=19000 : Creates your first separate data compartment inside that extended partition container with a size of roughly 19 GB.

```
DISKPART> create partition logical size=19000  
DiskPart succeeded in creating the specified partition.
```

- create partition logical size=19000 : Creates a second independent 19 GB data compartment right next to the first one.

```
DISKPART> create partition logical size=19000  
DiskPart succeeded in creating the specified partition.
```

- list volume: Prints out a structural summary of all your newly built drive partitions to verify they were created with the

proper sizes.

```
DISKPART> list volume
```

Volume ###	Ltr	Label	Fs	Type	Size	Status	Info
Volume 0	D	CCCOMA_X64F	UDF	DVD-ROM	5851 MB	Healthy	
Volume 1			RAW	Partition	20 GB	Healthy	
Volume 2			RAW	Partition	18 GB	Healthy	
* Volume 3			RAW	Partition	18 GB	Healthy	

- select volume 1 : Targets your first partition block.

```
DISKPART> select volume 1
```

Volume 1 is the selected volume.

- format fs=ntfs quick : Instantly writes a fresh, clean NTFS file system onto that volume so Windows can read and write files to it.

```
DISKPART> format fs=ntfs quick
```

100 percent completed

DiskPart successfully formatted the volume.

- assign letter A : Forcefully maps the specific drive letter A: to this volume.

```
DISKPART> assign letter A
```

DiskPart successfully assigned the drive letter or mount point.

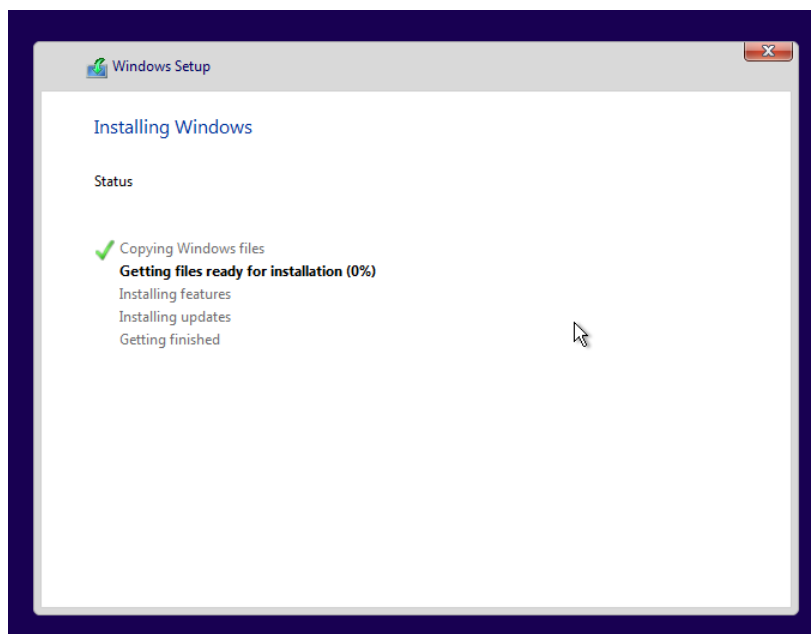
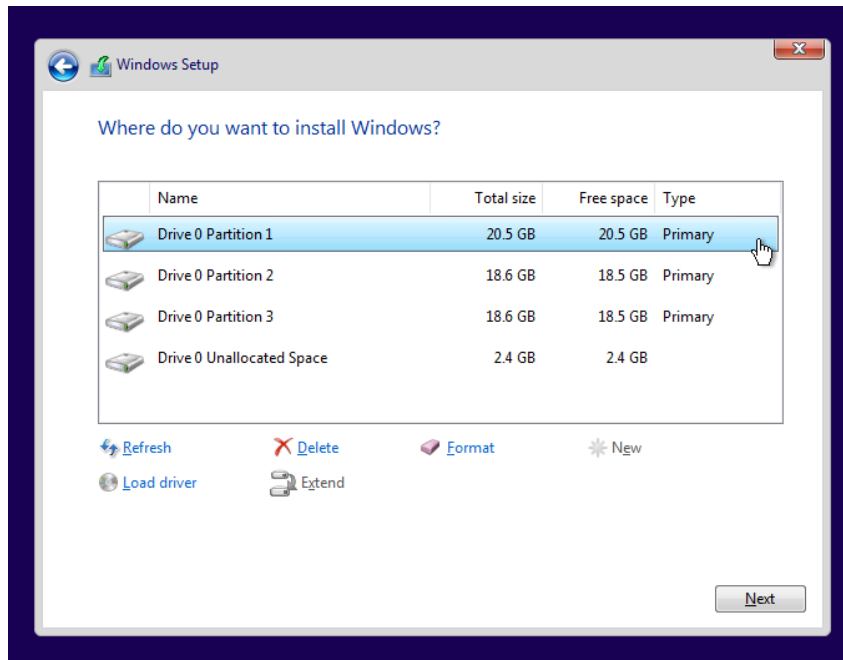
- Repeating for all volumes
select volume 2, format fs=ntfs quick, assign letter B
select volume 3, format fs=ntfs quick, assign letter C

```
DISKPART> list volume
```

Volume ###	Ltr	Label	Fs	Type	Size	Status	Info
Volume 0	D	CCCOMA_X64F	UDF	DVD-ROM	5851 MB	Healthy	
Volume 1	A		NTFS	Partition	20 GB	Healthy	
Volume 2	B		NTFS	Partition	18 GB	Healthy	
* Volume 3	C		NTFS	Partition	18 GB	Healthy	

E. Complete Windows Installation

Continue with installation -> select custom (advanced) -> select partition to install in



Who's going to use this PC?

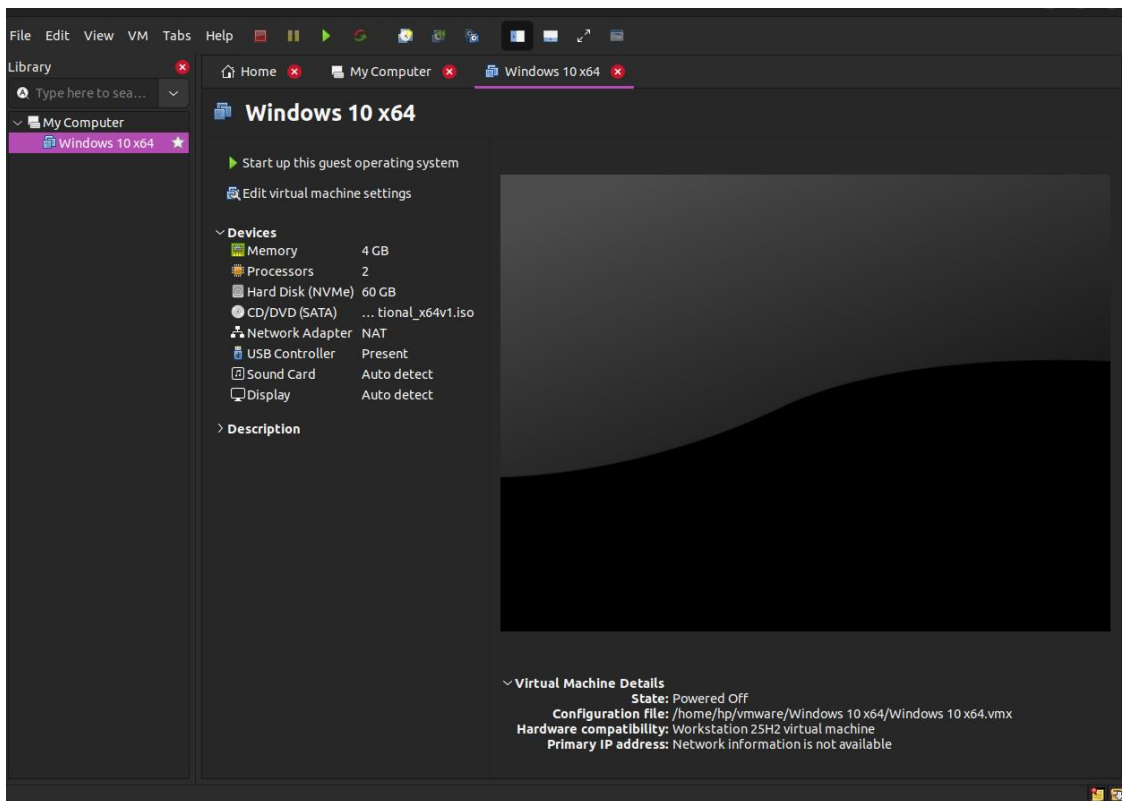
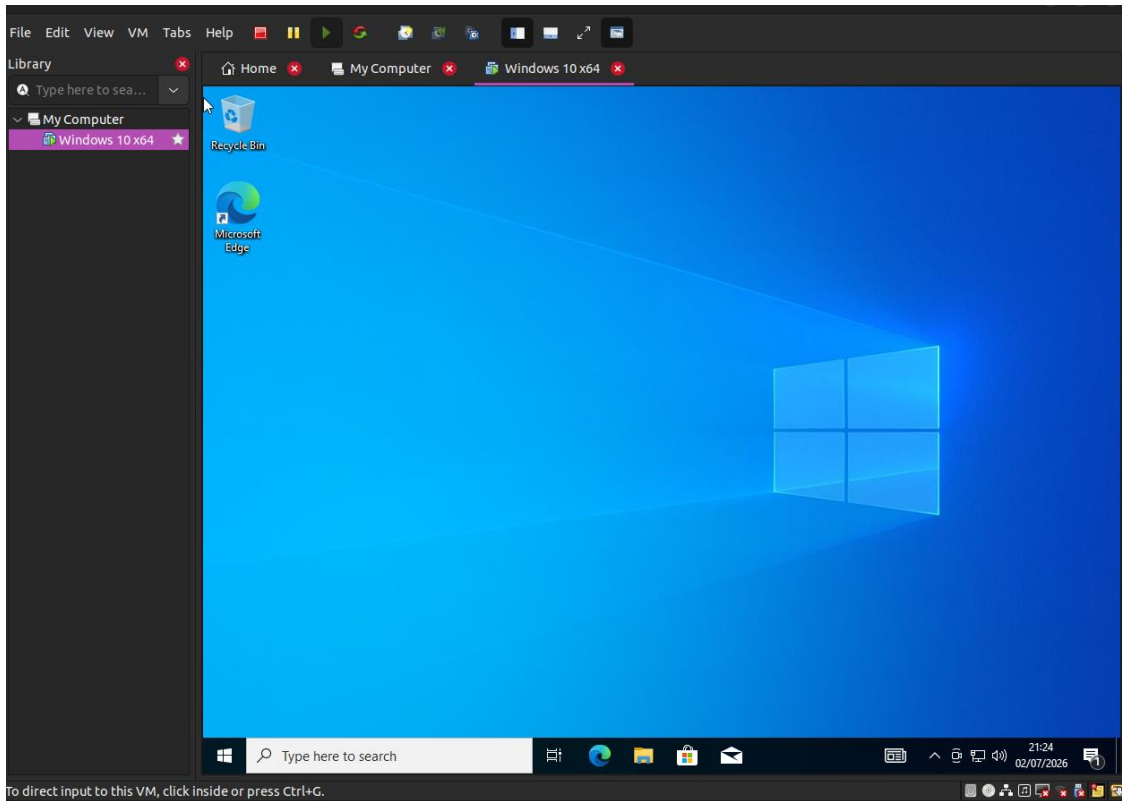
What name do you want to use?



Create a really memorable password

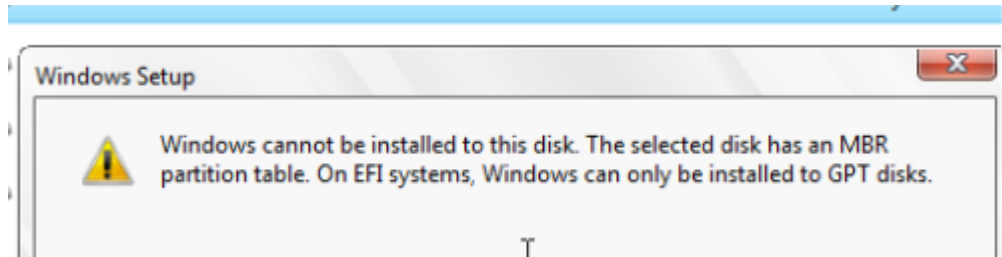
Make sure that you pick something you'll absolutely remember.





Challenges faced

1. Windows not being installed on disk because it's an MBR partition



2. Unability to perform command “create partition extended” on GPT partition

```
DISKPART> create partition extended

Logical and extended partitions cannot be created on a GPT disk.
Instead create primary partitions on a GPT disk.
```

Learning outcomes

- Creating a virtual machine
- Installing Windows 10
- Managing disk partitions using the Windows Command Line (DiskPart)
- Understanding Disk Partitioning Standards (MBR vs. GPT)
- Aligning Motherboard Firmware with Storage Layouts (UEFI vs. BIOS)

Conclusion

This document provides a simple step-by-step guide to building a professional virtual lab.

It spans environment preparation, virtual machine creation, storage initialization, firmware and partition alignment, and OS deployment.

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